

SingleRAN

Configuration Management Feature Parameter Description

Issue	01
Date	2024-03-09



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....1

1.1 SRAN20.1 01 (2024-03-09).....1

2 About This Document.....2

2.1 General Statements.....2

2.2 Applicable RAT.....3

2.3 Features in This Document.....3

3 Configuration Management.....4

3.1 Overview.....4

3.1.1 Definition.....4

3.1.2 Application Scenarios.....4

3.1.3 Architecture.....4

3.1.3.1 NEL.....5

3.1.3.2 EMS.....6

3.1.3.3 NMS.....6

3.1.4 Configuration Security.....6

3.2 Technical Description.....6

3.2.1 Network Deployment.....6

3.2.2 Network Adjustment.....8

3.2.3 Feature Activation.....11

3.2.4 Network Optimization.....11

3.2.5 Network Maintenance.....12

3.3 Related Documentation.....13

4 FAQ.....15

5 Glossary.....17

6 Reference Documents.....18

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN20.1 01 (2024-03-09)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve the optimal gains, contact Huawei professional service engineers.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial

features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview Feature Parameter Description*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
SRAN	MRFD-210301	Configuration Management	3.1 Overview
LTE FDD	LBFD-004007	Configuration Management	
NB-IoT	MLBFD-12000407	Configuration Management	
LTE TDD	TDLBFD-004007	Configuration Management	
NR	FBFD-010025	Basic O&M Package	

3 Configuration Management

3.1 Overview

3.1.1 Definition

Configuration management allows users to manage configuration data of network devices. This controls the running status of network devices and provides effective network O&M.

3.1.2 Application Scenarios

A configuration management system is required throughout the network's entire O&M cycle to allow users to perform the following:

- During network deployment, users can initialize configuration data and configure network device data.
- During network adjustment and optimization, a configuration management system allows users to rapidly and accurately modify configuration data for scenarios such as new feature activation, network capacity expansion, transmission reconstruction, base station migration, and wireless network performance optimization.
- During network maintenance, users can monitor, compare, analyze, and modify configuration data.

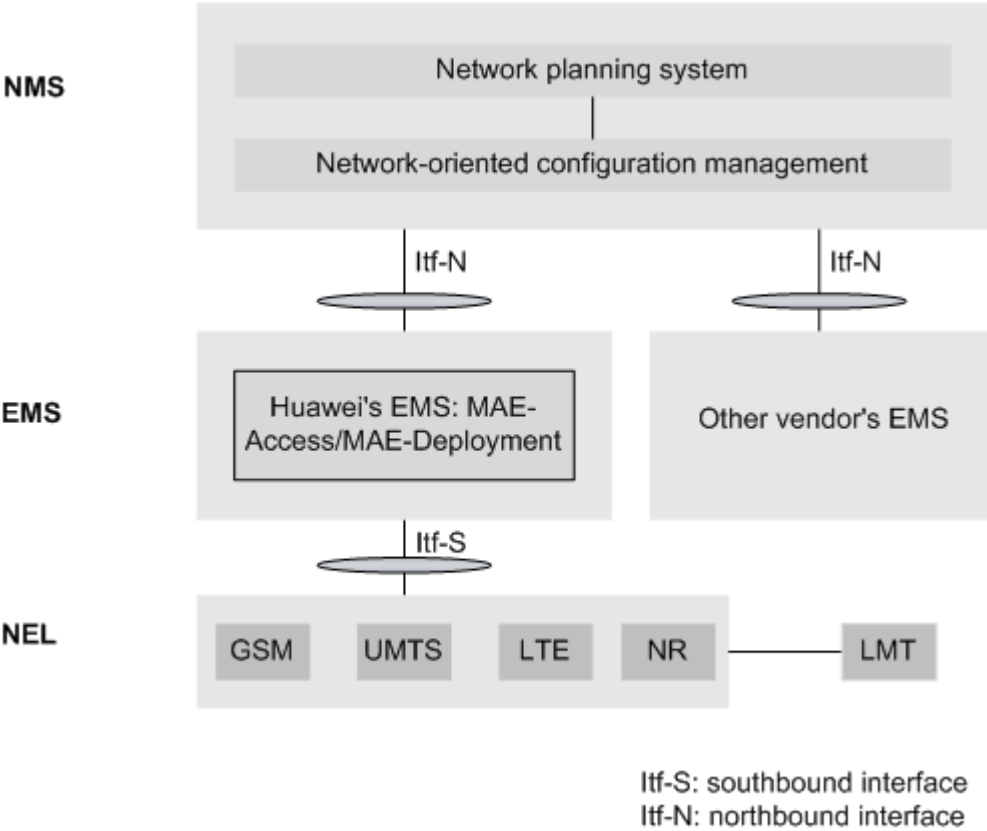
3.1.3 Architecture

Radio access networks (RANs) are divided into three layers:

- Network element layer (NEL)
- Element management layer (EML)
- Network management layer (NML)

Accordingly, as shown in [Figure 3-1](#), the configuration management architecture consists of three layers.

Figure 3-1 Configuration management architecture



3.1.3.1 NEL

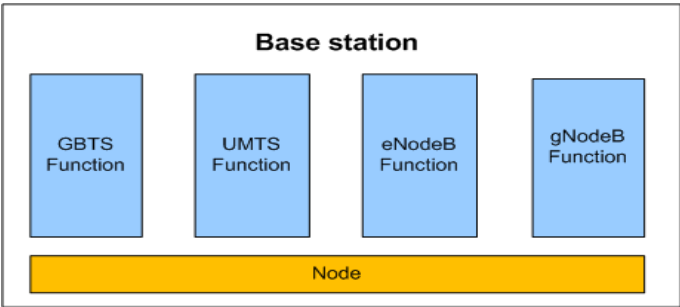
NEL allows users to modify single-NE configuration parameters through a man-machine language (MML) interface of the local maintenance terminal (LMT), an MML client integrated with the MAE, or the MAE-Deployment integrated with the MAE.

- Base stations provide both RAN non-real-time (RAN-NRT) and RAN real-time (RAN-RT) functions.

Base Station Deployment

Figure 3-2 shows the configuration model. Base stations provide a unified O&M channel for configuration management.

Figure 3-2 NEs in the traditional base station deployment



3.1.3.2 EMS

EMS allows users to modify and manage single-subnet configuration parameters. Huawei provides the professional configuration solution MAE-Deployment to implement EMS-layer configuration management. The MAE-Deployment centrally manages a SingleRAN network and provides man-machine interaction wizards and scenario-based designs to efficiently adjust networks in complex scenarios. It also provides the northbound interface (Itf-N) for integration with an upper-layer NMS. In principle, all configuration operations are recommended to be performed on the MAE-Deployment.

NOTE

The Itf-N is an open interface between the Huawei's EMS and a telecom operator's upper-layer NMS.

3.1.3.3 NMS

NMS involves configuration management for the entire network with a focus on connection to multiple vendors' EMSs. Huawei MAE-Deployment is connected to a telecom operator's NMS through the Itf-N. An NMS usually has a network planning system. After making parameter modification decisions, the network planning system sends parameter modification information to the MAE-Deployment through the Itf-N. The MAE-Deployment then modifies parameters.

3.1.4 Configuration Security

Configuration management provides a series of security measures to protect configuration data. For details, see *OM Security*.

3.2 Technical Description

3.2.1 Network Deployment

Introduction

Configuration management in this scenario specifies configuring data for newly deployed base station controllers or base stations.

For example, the deployment of new NodeBs or eNodeBs is further divided into the following three sub-scenarios according to the quantity of new base stations and configuration data differences between the new and existing base stations:

- Either a small number of base stations will be deployed or no base station has been deployed on the live network.
- A large number of base stations will be deployed and significant differences exist between new and existing base station configuration data.
- A large number of base stations will be deployed and slight differences exist between new and existing base station configuration data.

Recommended Configuration Functions

To deploy a new base station controller, users can perform MML batch configuration on the LMT or MAE. Alternatively, users can configure data using the MAE-Deployment general configuration window, device panel, and transport- and radio-layer planning data.

To deploy a new base station, it is recommended that the MAE-Deployment be used. [Table 3-1](#) describes the recommended configuration functions.

Table 3-1 Recommended configuration functions for new base station deployment

Scenario	Configuration Function	Description
Either a small number of base stations will be deployed or no base station has been deployed on the live network.	Single base station deployment using the wizard	Create the first base station and use its configuration data as a template for batch deployment of the remaining base stations.
A large number of base stations will be deployed and significant differences exist between new and existing base station configuration data.	Batch base station deployment by customizing a summary data file	Use the template for identical configuration data among base stations and customize configuration data that is different among the base stations.
A large number of base stations will be deployed and slight differences exist between new and existing base station configuration data.	Batch base station deployment using a summary data file based on the exported and customized managed object class (MOC)	Apply the sample base station data for identical configuration data among base stations and manually customize configuration data that is different among base stations.

Process

[Figure 3-3](#) shows the single base station deployment process using the wizard.

Figure 3-3 Single base station deployment using the wizard

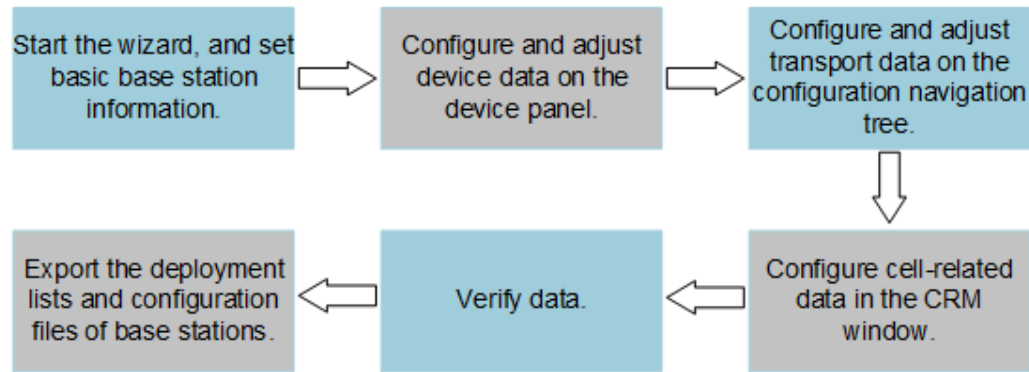


Figure 3-4 shows the batch base station deployment process by customizing a summary data file.

Figure 3-4 Batch base station deployment by customizing a summary data file

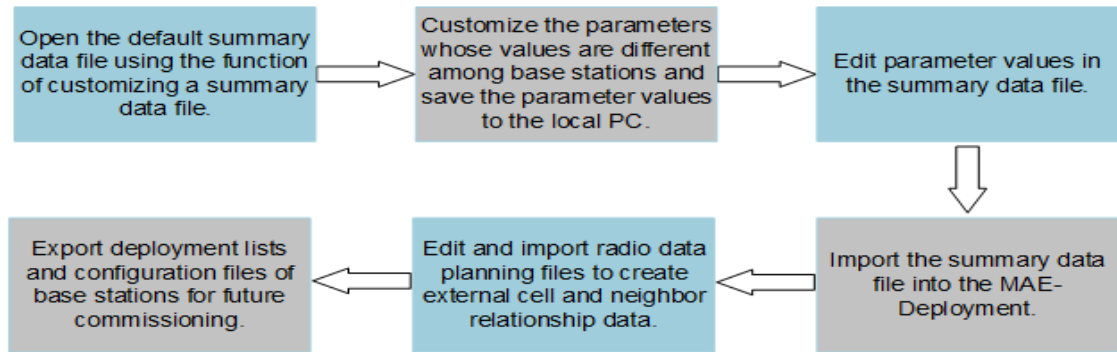
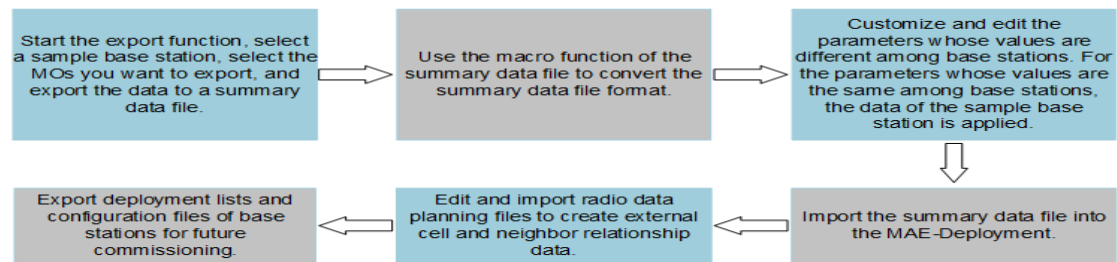


Figure 3-5 shows the batch base station deployment process using a summary data file based on the exported and customized MOC.

Figure 3-5 Batch base station deployment using a summary data file based on the exported and customized MOC



3.2.2 Network Adjustment

Introduction

Configuration management in this scenario specifies adjusting networks in which services are not evenly distributed. Network adjustment involves the following scenarios:

- Device capacity expansion
For example, adding a service processing board or interface board for an existing base station or base station controller.
- Transmission adjustment
This includes transmission capacity expansion and transmission reconstruction. For example, adjusting the clock unit and virtual local area network (VLAN).
- Cell or carrier capacity expansion
For example, changing the NodeB configuration from 3x1 to 3x2.
- Base station reconstruction
The typical scenarios include radio access technology (RAT) addition and co-MPT base station reconstruction.
 - RAT addition refers to adding new RATs to a co-MPT base station. For example, add LTE to a co-MPT G*U base station to make it a co-MPT G*U*L base station.
 - Co-MPT base station reconstruction refers to changing a separate-MPT base station to a co-MPT base station. For example, change a separate-MPT G&U base station to a co-MPT G*U base station.
- Other complex scenarios
Complex network adjustment scenarios may simultaneously involve device, transport, and radio data adjustment, such as base station migration, BSC node redundancy, and RNC pool configuration.

Recommended Configuration Functions

Table 3-2 describes the recommended configuration functions for network adjustment.

Table 3-2 Recommended functions for network adjustment

Scenario	Configuration Function	Description
Device capacity expansion	Base stations: Import and export of batch configuration data on the MAE-Deployment Base station controllers: Use the MAE-Deployment general configuration window and the device panel	-

Scenario	Configuration Function	Description
Transmission adjustment	Base stations: Import and export of batch configuration data on the MAE-Deployment Base station controllers: Import and export of Abis or lub batch configuration data on the MAE-Deployment	The BSC6960 does not support import and export of Abis or lub batch configuration data, which is incorporated into the import and export of base station controller configuration data function.
Cell or carrier capacity expansion	Single base station: Radio data configuration using the carrier resource management (CRM) Batch base stations: Import and export of adjusted radio frequency data	-
RAT addition	Single base station: NE shortcut menu on the MAE-Deployment graphical user interface (GUI) Batch base stations: Customized summary data files using the MAE-Deployment as well as import and export of batch configuration data	-
Co-MPT base station reconstruction	Data configuration conversion wizard for co-MPT base station reconstruction on the MAE-Deployment	Co-MPT base station reconstruction involves device, transport, and radio data conversion. Radio data reconstruction only involves radio, remote radio unit (RRU), and baseband processing unit (BBP) data conversion.
Base station migration	MAE-Deployment migration wizard	During a base station migration within a BSC/RNC or across BSCs/RNCs, users only need to plan transport parameters and certain radio parameters. The MAE-Deployment will automatically adjust the relations among the source BSCs/RNCs, destination BSCs/RNCs, and neighboring NEs. No manual operation is required.

Scenario	Configuration Function	Description
BSC node redundancy	BSC node redundancy configuration on the MAE-Deployment	The MAE-Deployment supports batch changes from single-homed GBTSS/eGBTSSs to dual-homed GBTSS/eGBTSSs or dual-homed GBTSS/eGBTSSs to single-homed GBTSS/eGBTSSs and data checks.
RNC pool configuration	RNC pool configuration on the MAE-Deployment	The MAE-Deployment supports batch changes from single-homed NodeBs to dual-homed NodeBs or dual-homed NodeBs to single-homed NodeBs and data checks. The BSC6960 does not support the RNC pool function.

3.2.3 Feature Activation

Introduction

Configuration management in this scenario specifies configuring data to activate a new feature on a legacy network, such as, the Carrier Aggregation feature after the deployment of an eNodeB.

Recommended Configuration Functions

Recommended configuration functions on the MAE-Deployment include feature status query, batch feature activation, and validity checks. For details about feature operation and maintenance of the MAE-Deployment, see **Operation and Maintenance > MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance** in the MAE product documentation.

3.2.4 Network Optimization

Introduction

Configuration management in this scenario specifies adjusting parameters of a running network to provide better network services. Compared with network adjustment, network optimization focuses more on radio-layer parameter adjustment.

The MAE-Deployment supports the following tasks for network optimization:

- Batch adjustment of frequencies or scrambling codes in a cell
- Batch adjustment of neighboring cells or neighboring cells' BCCH allocation (BA) data
- Batch algorithm parameter adjustment for cells, carriers, and channels

- Data reconfiguration of certain service features, such as reconstructing a cell that uses High Speed Packet Access (HSPA) and changing the operator to which a cell belongs in a multi-operator core network (MOCN)

Recommended Configuration Functions

Table 3-3 describes multiple batch data modification methods using the MAE-Deployment.

Table 3-3 Recommended functions for network optimization

Configuration Function	Description	Application Scenario
Importing data from data planning files	To quickly adjust data in batches, export a summary data file, modify the data, and import the file.	This method is applicable if users configure different values for multiple objects. The MAE-Deployment provides multiple data planning files for different network optimization tasks.
Modifying data in batches by using the wizard	Select some or all of the configuration parameters for an object as a reference and apply these parameters to another object.	This method is applicable if users configure parameters to the same value for a single type of objects.
Reconfiguring data for some service features	The MAE-Deployment provides a configuration wizard that supports one-click configuration for an object.	N/A
Importing data from XML files through the northbound interface	After the data is configured, the MAE-Deployment exports configuration data files in either MML or XML format for each NE. Users can use the script executor provided by the MAE-Deployment to remotely deliver and activate the configuration data files to each NE through the MAE.	This method is applicable if the MAE-Deployment is connected to the 3GPP R6-compliant northbound interface of the telecom operator's NMS.

3.2.5 Network Maintenance

Introduction

Configuration management in this scenario specifies implementing routine inspection on configuration parameters for a running network, including

parameter browsing, checks, and analysis. Configuration parameters are not modified and the services on the live network are not affected.

Recommended Configuration Functions

Table 3-4 describes the recommended configuration functions for network maintenance.

Table 3-4 Recommended configuration functions for network maintenance

Scenario	Configur ation Function	Description
Paramete r browsing	Configura tion reports	The MAE-Deployment provides configuration reports for various scenarios during network maintenance. These reports can be saved in multiple reports as required and enable users to view data for different versions of multiple NEs on the live network. The MAE-Deployment also allows for statistics collection on the entire network.
Paramete r check	Consisten cy check	A network generates a large amount of data and the correlations between configuration objects are complex. The MAE-Deployment checks the integrity, redundancy and consistency of configuration data, and generates an error report. It also automatically recovers certain data according to check results.
Paramete r analysis	Paramete r comparis on	During the initial phase of network construction, system default values are used for certain algorithm parameters. During network maintenance and optimization, users can reconfigure algorithm parameters as required. The MAE-Deployment provides comparison between the actual values of the algorithm parameters for a base station, controller, or cell with the default values or configuration baseline data for subsequent adjustment.
	Paramete r filter	The MAE-Deployment provides default parameter sets, also known as parameter filters, according to different NEs and versions to help users quickly select the parameters to be compared. In addition, users can customize parameter filters.

3.3 Related Documentation

The NE configuration management documentation is as follows:

- Product documentation: Initial Configuration Guide, Reconfiguration Guide, MML Command Reference, and MO and Parameter Reference
- Feature documentation: Feature Parameter Description
- Release documentation: MML Command Changes, MO and Parameter Changes, Reserved Parameter List, and Disuse Parameter List
- Northbound documentation: Northbound Parameter Changes and Northbound Parameter List

The configuration management tool documentation is as follows:

- LMT: LMT User Guide
- MAE: MAE MBB Network Management System Product Documentation
- MAE-Deployment: MAE-Deployment Product Documentation

The preceding documentation can be obtained from <https://support.huawei.com>.

4 FAQ

Definition

A parameter involves attributes of an NE configuration object, while a parameter value is configured for each attribute.

Each parameter has multiple fields, such as **Parameter ID**, **Parameter Name**, **NE**, **MML Command**, **Meaning**, and **IsKey**. For detailed attribute lists, see parameter references.

Setting Modes

- Manual settings: Parameter values are manually planned and adjusted.
- Automatic setting: Configuration data is automatically generated. For example, boards are automatically added after addition of a subrack, or parameter values are automatically generated using the self-organizing network (SON).

Branch Parameters

Branch parameters take effect only when certain conditions are met. If certain conditions are not met, a system cannot obtain a parameter value and the branch parameter does not take effect. **Parameter Relationship** in parameter references provides parameter relationship with conditions.

Mandatory attributes of a branch parameter are configured according to the following principles:

- If a branch parameter is mandatory and in a default branch relationship, **Mandatory** is set to **YES**.
- Otherwise, **Mandatory** is set to **NO**.

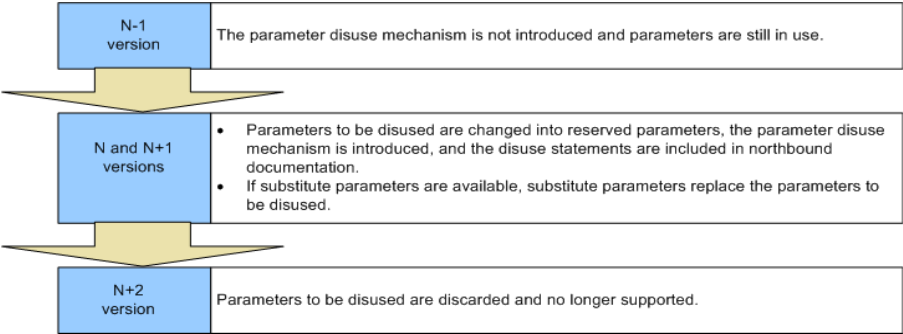
NOTE

For example, if a branch parameter is not in a default branch relationship, **Mandatory** is set to **NO** regardless that the parameter is mandatory under the branch relationship.

Northbound Parameter Disuse

The status of northbound parameters during a disuse period (N to N+2 versions) is shown in [Figure 4-1](#).

Figure 4-1 Northbound parameter disuse



5 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

6 Reference Documents

None